

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
Electrical Engg. Department

Mid Term 2
Btech 4th sem
Control Systems (R&A)
RAPC208

1. Consider

$$F(s) = s^6 + 4s^5 + 12s^4 + 16s^3 + 41s^2 + 36s + 72$$

Find the number of roots lying in the right half plane (if any). (4)

2. Draw the root loci of the open-loop transfer function of the feedback control system. (7)

$$G(s) H(s) = \frac{K}{s(s+4)(s+5)}$$

3. For the control system shown below, determine the steady-state error resulting from the input.

$$r(t) = (5 + 8t + 3t^2) U(t) \quad (4)$$

